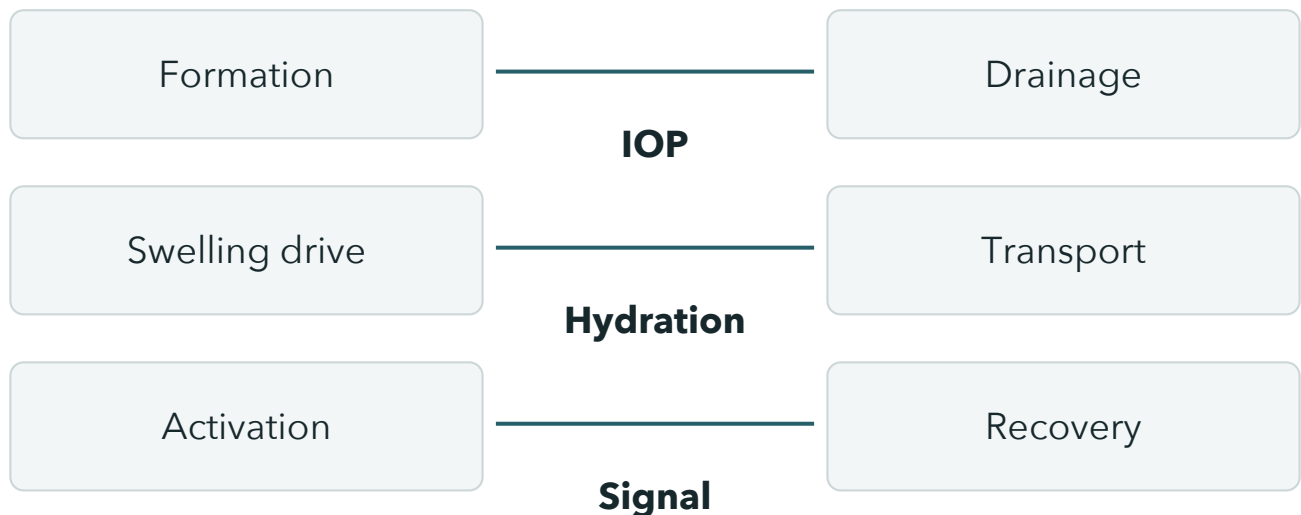


Ocular physiology and molecular function

Part 1 | Physiology is an interacting set of balances

Flow, transport and signalling can change independently despite a similar final phenotype.



Three balances connect secretion/drainage, hydration and signalling activation/recovery.

Compare the variables

IOP reflects formation, drainage resistance and downstream pressure. Corneal thickness reflects fluid transport and stromal swelling tendency. A photoreceptor response reflects activation followed by recovery. A single endpoint cannot identify every parameter in these systems; a controlled perturbation can separate their contributions.

Aqueous pressure has more than one determinant

The simplified conventional relation is useful when its excluded routes are stated.

$$P=(F-U)/C+P_e$$

Parameter	Increase, others fixed	Model IOP
Formation F	More conventional flow demand	Rises
Pressure-insensitive outflow U	Less conventional flow demand	Falls
Facility C	Less gradient needed	Falls
Downstream P_e	Higher baseline	Rises

Measurement subtlety

Pressure-decay facility can be overestimated if pressure elevation also reduces inflow and the analysis assigns the entire net loss to outflow. This pseudofacility contribution is an assumption problem, distinct from calculating the pressure of a fixed-parameter system.

Membrane permeability sets effective tonicity

Measured osmoles influence sustained volume only if they maintain a transmembrane gradient.

Equal added measured osmolality

Solute	After equilibration	Persistent effect
Freely permeant	Distributes across membrane	Gradient partly dissipates
Effectively extracellular	Stays outside during experiment	Water loss persists

Why water moves

Water redistributes according to effective osmotic gradients. A permeant solute can produce a transient effect before equilibrating, so sampling time matters. The comparison assumes no active volume-regulatory response; living cells may subsequently alter their own solute content and depart from the simple model.

Retinal circuits encode changes at several levels

Photoreceptor output and postsynaptic response polarity are not identical.

During illumination

Level	Primary change
Photoreceptor	cGMP-gated current falls; glutamate falls
ON bipolar	Sign-inverting pathway depolarises
OFF bipolar	Excitatory glutamate drive falls
Ganglion receptive field	Centre and surround weight spatial input

Activation and recovery

Rhodopsin shut-off, transducin-PDE deactivation and cGMP restoration terminate different active intermediates. Normal onset with a prolonged response therefore favours a recovery process over failed initial detection. It does not uniquely identify one gene without another discriminating measurement or rescue.

Optical clarity depends on molecular organisation

Similar opacity can arise through hydration, matrix disorder or protein aggregation.

Tissue

Feature	Mechanism	Distinct readout
Corneal stroma	Hydration/fibril organisation	Thickness and scatter
Lens cells	Polyol osmotic/redox effects	Volume and glutathione
Lens proteins	Crystallin aggregation	Solubility and scatter

Convergent consequences

The same visual symptom can reflect distinct physiological levels. A normal lens curvature does not exclude increased scatter; a normal collagen amount does not exclude disturbed organisation. Preserved gross structure constrains a mechanism but does not prove preserved molecular function.